

CareerPath: A Structured Career Guidance Platform for students

A Project Report submitted by
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in partial fulfilment of the requirements for the award of the degree of
M.Tech.



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Declaration

I hereby declare that the work presented in this Project Report titled “CareerPath: A Structured Career Guidance Platform for students”– M.Tech (AI/ML). submitted to the Indian Institute of Technology Jodhpur in partial fulfilment of the requirements for the award of the degree of M.Tech. (AI/ML) is a bonafide record of the research work carried out under the supervision of Professor Dr. Dinesh Mohan Joshi. The contents of this Project Report in full or in parts, have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.

Signature

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Certificate

This is to certify that the Project Report titled CareerPath: A Structured Career Guidance Platform for students, submitted by Manvendra Pratap Singh (M25AI2122) to the Indian Institute of Technology Jodhpur for the award of the degree of M.Tech. (AI/ML), is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature

Dr. Dinesh Mohan Joshi

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Abstract

CareerPath is an excellent career guidance platform designed to assist students in making decisions regarding their education and future professional career. It is specifically developed for the students of villages and small towns regions where career-related information is less. Instead of just internet searches, this platform presents various academic streams and their internal career paths in a direct, clear, and easy manner.

The platform has three primary parts: a mobile app for students, a web app for management, and a backend system for API. By mobile app, students can access information of academic streams and career path, with resources such as books, colleges, and job trends. The web app enables administrators to add and update data, while the backend API ensures the smooth and efficient communication..

In this report I explain my motivation, problem statement and objective in simple language.

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Background

After completing the 10th and 12th exams, students are so confused while selecting their academic path and career. Mostly in the tier 2 and 3 cities and villages and, where access to proper career guidance are either limited. Career decisions depends on a word of mouth of uncle and senior cousin and the family's financial situation. In reality, most students ends up choosing a career based without complete informations.

While modern AI tools and search engines can certainly provides answers to specific queries, they operate solely based on what the student already knows and ask about. This creates a sort of "discovery gap", meaning that if a student is completely unaware of a particular career path. CareerPath has been designed to bridge this very gap by present career-related information in a structured and user-friendly manner, this enabling students to explore and understand various options, evn they know about that path and then they take good decisions.

Problem Statement

The problem is very straight foreword.

- Students often face difficulties in choosing their careers after completing the 10th and 12th grades.
- This problem is more in rural, semi-urban, and other regions where statmic career guidance is not available.
- Career-related decisions depend on several factors, such as:
 - Stream of student
 - Options for that stream
 - Educational institutions
 - Employment opportunities
 - Study resources
 - Financial condition

In reality, students make decisions based on incomplete or limited information.

To address this issue, CareerPath presents career-related information within a structured and systematic model.

Aim and objectives

Primary Aim

To develop a structured career guidance platform wherein academic and career-related information is organized within a systematic system. Then deliver this information to students through a simple, accessible, and user-friendly mobile application.

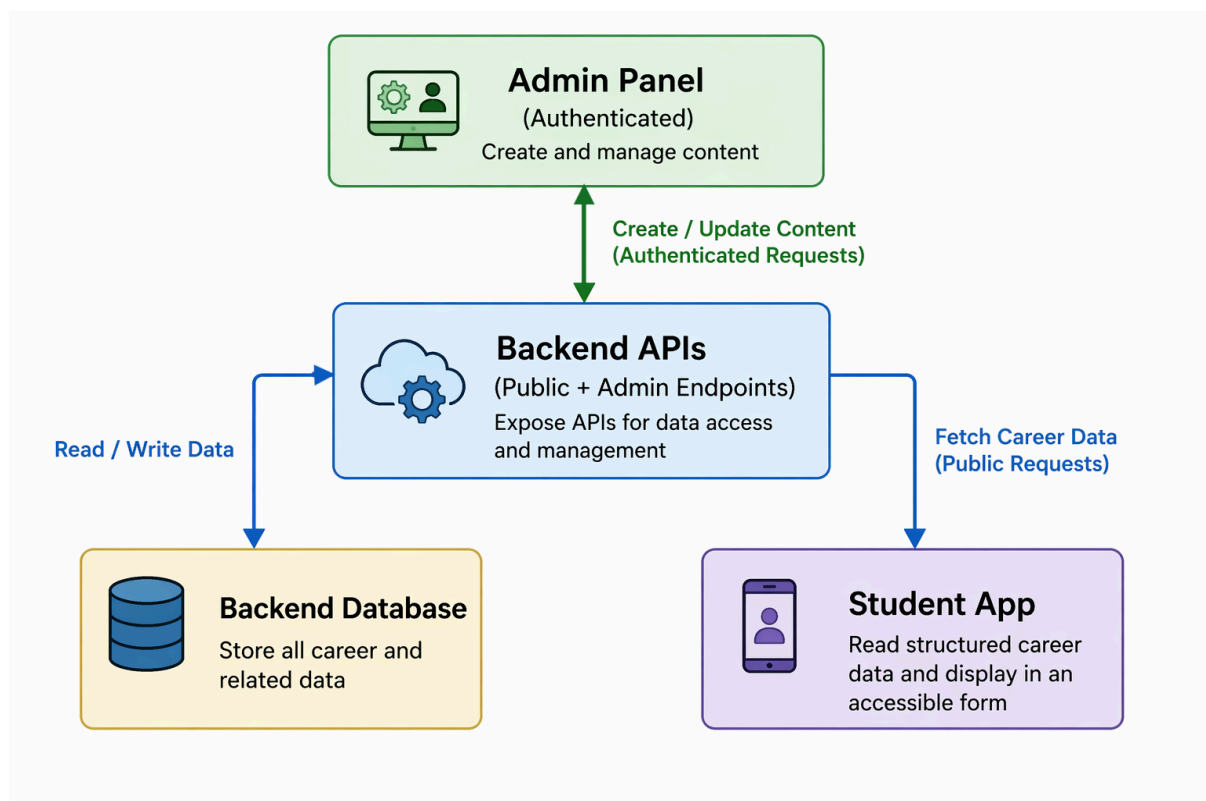
Specific Objectives

- To create a structured database of career paths categorized academic streams such as Science, Commerce, and Arts.
- To provide students with guided navigation, ranging from broad academic streams to specific career options.
- To make supporting resources available at the granular level (Leaf Nodes), such as:
 - Books
 - Educational Institutions
 - Employment and Job Sectors
- To develop an Admin Panel for streams, branches, books, institutions, career paths, and other related data without requiring modifications to the source code.
- To centrally manage all data through backend APIs and database services.
- To ensure the platform remains extensible, allowing for the future integration of new features such as recommendation systems, search capabilities, and analytics.

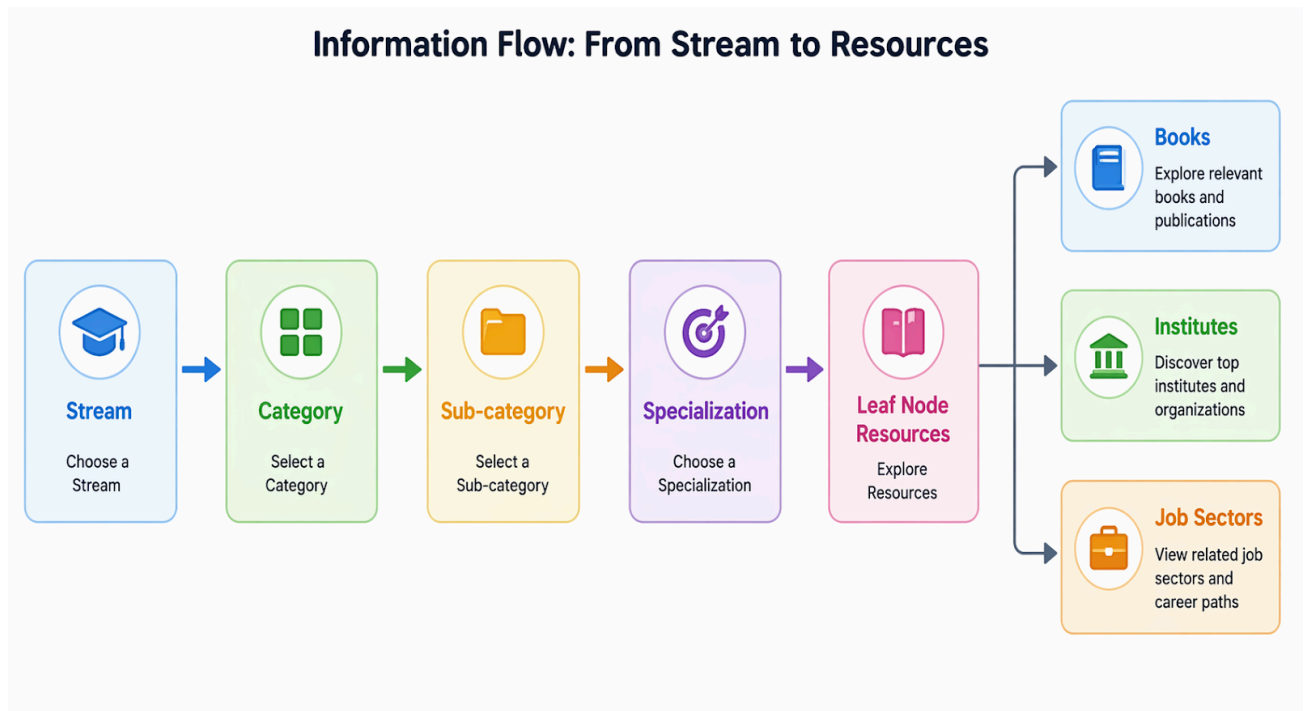
Architecture

Conceptual Flow

- Data is stored in the backend database.
- Backend APIs expose public and admin endpoints.
- The admin panel uses authenticated APIs to create and manage content.
- The student app reads structured career data and displays it in an accessible form.



Conceptual Flow Hierarchical Data Model



Technology Stack

Layer	Technology
Student App	Flutter / Dart
Admin Panel	React JS
API	Python
Database	SQLite

Key Features

Features for Students

- Onboarding and profile creation
- Stream wise suggestions
- Explore tab for full career path
- Struture navigation with all screens
- Leaf detail view with resources
- Search and bookmarks
- Offline-friendly with local data in app

Features for Administrators

- Login for admin access
- CRUD management for streams and nodes
- Resource management for books, institutes, sectors, and exams
- API data control without developer intervention

Features for the Backend

- Public read APIs
- Admin write APIs
- Database handling

Delivered Items

- APP for student/public open: A public-facing application for students to explore structured career paths and related resources.
- Admin panel: Managed by admin to add streams, branches, books, institutes, paths, and other platform features.
- Backend: All database handling, API services, and application running support.

Application Link: [Playstore](#)

Conclusion

CareerPath is a practical platform for making career guidance accessible to all students. By using career-related information in a simple and structured format, it shows students in understanding and selecting their options based on correct information. It has a mobile app, an admin panel, and an API backend, with future scalability.

This project shows that a structured software platform can address real-world challenges in the education sector. In the future, CareerPath can be evolved into a student guidance system by adding advanced search capabilities, interest filtering, analytics, and AI recommendation.